Course Outline for EE-184

Part A

1. Course Code: EE-184

2. Course Title: Electronic Devices and Circuits (Sessional)

3. Course Type: Core Course

4.Level/ Term: Level: 1 Term: II

5. Academic Session: 2020-21

6. Course Teacher: Annesha Das, Assistant Professor, Dept. of CSE, CUET and
Ashim Dey, Assistant Professor, Dept. of CSE, CUET

7. Prerequisite(s): None

8. Credits: 1.5 (3 hours of lab work per week)

9. Contact Hours: 3 hours of lab work per week

10. Total Marks: 150

11. Rational of the Course:

This course will cover the design of different digital electronics circuits and different pulse generations circuits. This course is important to understand how linear and nonlinear wave shaping circuits; pulse generating circuit such as multivibrator; oscillator; amplifier works. This a required course for all the students enrolling B. Sc. Engg. in CSE program. The catalogue description of the course is

Course Content:

Sessional based on the following topics:

Introduction to semiconductors; Bipolar Junction Transistor (BJT): *Bipolar Junction Transistor (BJT): principle of operation, I-V characteristics; Transistor circuit configurations (CE, CB, CC), BJT biasing, load lines; BJTs at low frequencies; Hybrid model, h parameters, simplified hybrid model; Small-signal analysis of single and multi-stage amplifiers, frequency response of BJT amplifier.* ***Field Effect Transistors (FET):*** *principle of operation of JFET and MOSFET; Depletion and enhancement type NMOS and PMOS; biasing of FETs; Low and high frequency models of FETs, Switching circuits using*

FETs; ***Introduction to CMOS; OPAMP:*** *linear applications of OPAMPs, gain, input and output impedances, active filters, frequency response and noise, Non-linear applications of OP AMP;* ***DAC-ADC:*** *A/D and D/A converters with applications; Sample and hold circuit;* ***LED-LCD:*** *Introduction of LED and LCD with application; Oscillators; Silicon Controlled Rectifiers (SCR), TRIAC, DIAC and UJT: characteristics and applications; Introduction to IC fabrication processes. Stabilizer and UPS.*

Wave shaping circuits: *Linear wave shaping; Non-linear wave shaping; Diode wave shaping technique s, clipping and clamping circuits (Diode applications); regulated power supply using zener diode; comparator circuits;* ***Multivibrators:*** *monostable, bistable and astable multivibratiors; Schmitt trigger; blocking oscillators and time-base circuit; Timing circuits, Simple voltage sweeps, linear current sweeps.*

12. Course Objectives:

(a) Familiarization with different Electronic Devices and laboratory equipment.
(b) Understand the theory and operation of the wave shaping circuits and analyze their responses.
(c) Understand and analyze the response of pulse generating circuits.
(d) Design and analyze different transistor configuration and observe their output curve.

13. Course Learning Outcomes (CLOs) and Mapping of CLOs with Program Learning Outcomes (PLOs)

a) CLOs

	Course Learning Outcomes (CLOs)	**Blooms Level (Optional)**
CLO1	Know about different Electronic Devices with their application and laboratory equipment with working.	C1
CLO2	Demonstrate the operation of the wave shaping circuits and analyze their responses.	C2,P3
CLO3	Demonstrate the operation of pulse generating circuits and analyze their response.	C2,P3
CLO4	Design different transistor configuration and analyze their output curve.	C3, P3

b) Mapping of CLOs with PLOs

No.	**(CLOs)**	**PLO 1**	**PLO 2**	**PLO 3**	**PLO 4**	**PLO 5**	**PLO 6**	**PLO 7**	**PLO 8**	**PLO 9**	**PLO1 0**	**PLO 11**	**PLO 12**

1	CLO1	X											
2	CLO2			X		X							
3	CLO3					X							
4	CLO4			X									

Part B

14. Course plan specifying content, CLOs, co-curricular activities (if any), teaching learning and assessment strategy mapped with CLOs

Course Plan

`	Topic	Teaching-Learning Methodology	Assessment Method	Corresponding CLOs
Week-01	Familiarization with the laboratory equipment for electronic devices & circuits.	• Lecture on theatrical background and design principle • Hands on demonstration on the implementation • Group Discussion	Report	• CLO-1
Week - 02	Study the response of RC High Pass Circuit	• Lecture on theatrical background and design principle • Hands on demonstration on the implementation • Group Discussion	Lab Performance Report	• CLO-2
Week - 03	Study the response of RC Low Pass Circuit	• Lecture on theatrical background and design principle • Hands on demonstration on the implementation • Group Discussion	Lab Performance Report	• CLO-2

Week - 04	Design and construction of Nonlinear wave shaping circuit: Shunt Clipping Circuits	• Lecture on theatrical background and design principle • Hands on demonstration on the implementation • Group Discussion	Lab Performance Report	• CLO-2
Week - 05	Design and construction of Nonlinear wave shaping circuit: Series Clipping Circuits	• Lecture on theatrical background and design principle • Hands on demonstration on the implementation • Group Discussion	Lab Performance Report	• CLO-2
Week - 06	Design and implementation of Nonlinear wave shaping circuit: Clamper Circuits (Positive and Negative)	• Lecture on theatrical background and design principle • Hands on demonstration on the implementation • Group Discussion	Lab Performance Report	• CLO-2
Week - 07	Construction and verification of Transistor Configuration : Common Base	• Lecture on theatrical background and design principle • Hands on demonstration on the implementation • Group Discussion	Lab Performance Report	• CLO-4
Week - 08	Construction and verification of Transistor Configuration : Common Emitter	• Lecture on theatrical background and design principle • Hands on demonstration on the implementation • Group Discussion	Lab Performance Report	• CLO-4
Week - 09	Construction and verification of Transistor Configuration :	• Lecture on theatrical background and design principle • Hands on demonstration on the implementation	Lab Performance Report	• CLO-4

	Common Collector	• Group Discussion		
Week - 10	Implementation of Astable Multivibrator Using IC 555 Timer	• Lecture on theatrical background and design principle • Hands on demonstration on the implementation • Group Discussion	Lab Performance Report	• CLO-3
Week - 11	Lab Test - 01	• Hands on demonstration on the implementation • Group Discussion	Lab Performance	
Week - 12	Lab Test - 02	• Hands on demonstration on the implementation • Group Discussion	Lab Performance	
Week - 13	Viva-voce Quiz			

Part C

15. Assessment and Evaluation

l) Assessment Strategy

Quizzes	15%
Viva-voce	15%
Class performance including reports	60%
Attendance	10%
Total	100%

2) Marks distribution:

a) Continuous Assessment: 70%
b) Summative: 30%

3) Make-up Procedures:

- Course teacher may arrange for makeup lab schedule if necessary.

Part D

16. Learning Materials

1) Recommended Readings

- A Textbook of Electrical Technology by B.L. Theraja (Volume IV)
- Handbook of Electronics Writer: Kumar Gupta
- Pulse, Digital and Switching Waveforms by Jacob Millman and Herbert Taub

2) Supplementary Readings

- Electronic Devices by Thomas L. Floyd
- Principles of Electronics by V.K. Mehta and Rohit Mehta
- Digital Systems: Principles and Applications by Ronald J. Tocci and Neal S. Widmer